Group -I crossection derivation

Corrections (blue = new, red = remove; see Group_check.pdf):

- **Row V, 2b UV (E1), Eqs. after "Setting $\Lambda \to 0$" and in the final answer:** the bracket is $\left[2\log\frac{k^+}{\Lambda}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}\right]$. Your exact bracket just before it has $2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}=2\log\frac{k^+}{\Lambda}$; there is no $\log\frac{\vee}{k^+}$ in Row V ($0<p^+<k^+$).
- Row II, d^3N_{2b}: x' is integrated, $\int_{w,w',x'}$; Row II part 1: a stray comma in $\int_{w,w',x,x'}$.
- Display only: in Row III d^3N_1 the $\Big\{$ is never closed; in Row V part 2 ("Integrating w.r.t. w", "This can be written as", d^3N_{2a} and its plus-prescription form) a $\Big\}$ closes nothing.
- Row V part 1: $\int_{x,z,y,x',y',w'}$ (comma).
- Not marked (notation): the logarithms of Row III are principal values, $\log|\cdot|$; the arguments change sign at $p^+=k^+\frac{(x-w)^2}{(x-z)^2}$.

1 Group I (with c.c.)

1.1 Row I [Done ✓]

Only 4 ρ's

1.2 Row II [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Big[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')+\mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\Big]\times\Big[U^{ab}(w)\mathbb{A}_i^{(3)\,b}(k^+,-w)-\bar{\mathbb{A}}_i^{(3)\,a}(k^+,-w)\Big]$$

We write it in two parts:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',}e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\mu^2_{\overline{MS}}}{4}\right]\right]\times\left[\frac{11}{3}+\log\left[\frac{\Lambda}{k^+}\right]-\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}-1\right]\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\Big]\tag{1.1}$$

Rewriting, we get

$$\frac{d^3N_{1\,\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',}e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\mu^2_{\overline{MS}}}{4}\right]\right]\times\left[\frac{11}{3}-2\log\left(\frac{k^+}{\Lambda}\right)+\log\left(1-\frac{k^+}{\vee}\right)\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\Big]\tag{1.2}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w',x'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y} e^{ik\cdot(w-y)}e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}$$
$$\left[\left[-\log\left(\frac{\vee}{\Lambda}\right)+\log\left[\frac{\vee}{k^+}\right]\right]\left[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\right]-2\,k^i\,\log\left(\frac{\vee}{\Lambda}\right)-2k^i\,\log\left(\frac{k^+p^2}{\vee\,k^2+k^+p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\right.$$
$$\left.\log\left[1+\left(\frac{\vee}{k^+}-1\right)\frac{k^2}{(k-p)^2}\right]-\frac{(p^2\,k^i+k^2\,p^i)}{(k-p)^2}\,\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]-k^i\,\log\left[\frac{\vee}{k^+}\right]\right]\times$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{hbc}\,U^{ah}(w)\,\left\{\rho^b(\mathbf{x}),\,\rho^c(\mathbf{y})\right\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\left\{\rho^d(\mathbf{x}),\,\rho^e(\mathbf{y})\right\}\right] \tag{1.3}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{d^3N_{2\,a}}{d^3k}+\frac{d^3N_{2\,b}}{d^3k} \tag{1.4}$$

$$\frac{d^3N_{2\,a}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{k^+}\right]+\log\left[\frac{k^+}{\Lambda}\right]\right]\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.$$
$$\left.-\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]$$
$$-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}-\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]$$
$$-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{abc}}{32\pi^4}\,\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}\right.$$
$$\left.-\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\right] \tag{1.5}$$

$$\frac{d^3N_{2\,b}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w',x'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y} e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{hbc}\,U^{ah}(w)\,\left\{\rho^b(x),\rho^c(y)\right\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\left\{\rho^d(x),\rho^e(y)\right\}\right] \tag{1.6}$$

where

$$R^i = -\,2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}\,p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right]-\frac{p^2k^i+k^2p^i}{(k-p)^2}\,\log\left(1-\frac{k^+}{\vee}\right). \tag{1.7}$$

1.3 Row III

[Done ✓]

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_z\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')+\mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times$$
$$\left[-\bar{\mathbb{B}}_{3ji}^{\dagger\,ba}(k^+,w;p^+,-z)\bar{\mathbb{A}}_j^{(1)\,b}(p^+,-z)+\bar{\mathbb{B}}_{3ji}^{\dagger\,ba}(k^+,w;p^+,-z)U^{bc}(z)\mathbb{A}_j^{(1)\,c}(p^+,-z)\right]$$

$$
\begin{aligned}
\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& \frac{ig^4\, f^{abd}}{8\pi^4} \int_{k^+ + \Lambda}^{\vee} \frac{dp^+}{2\pi} \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2} \frac{(p^+ + k^+)}{(k^+ - p^+)^2} \frac{1}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& + \frac{ig^4\, f^{abd}}{8\pi^4} \int_{\Lambda}^{k^+ - \Lambda} \frac{dp^+}{2\pi} \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2} \frac{(p^+ + k^+)}{(k^+ - p^+)^2} \frac{1}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{8\pi^4} \int_{\Lambda}^{\vee} \frac{dp^+}{2\pi} \frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2} \frac{\big[k^+\,(x-w)^2 + p^+\,(x-z)^2\big]}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]} \frac{1}{p^+\,k^+} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{4\pi^4} \frac{1}{k^+} \int_{\Lambda + k^+}^{\vee} \frac{dp^+}{2\pi} \int_{x,y,z,x'} \frac{(x'-w')^i\,(y-z)^j\,(z-w)^k}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \frac{[k^+\,(x-w)^m - p^+\,(x-z)^m]}{[k^+\,(x-w)^2 - p^+\,(x-z)^2]} \frac{1}{(k^+ - p^+)} \\
& \left[\delta_{ik}\,\delta_{jm} - \frac{k^+}{p^+}\,\delta_{jk}\,\delta_{im} - \frac{k^+}{(k^+ - p^+)}\,\delta_{ij}\,\delta_{km}\right] \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \\
& \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{4\pi^4} \int_{\Lambda}^{k^+ - \Lambda} \frac{dp^+}{2\pi} \int_{x,y,z,x'} \frac{(x'-w')^i\,(y-z)^j\,(z-w)^k}{(x'-w')^2\,(y-z)^2\,\big[k^+\,(x-w) - p^+\,(x-z)\big]^2} \frac{[k^+\,(x-w)^m - p^+\,(x-z)^m]}{[k^+\,(x-w)^2 - p^+\,(x-z)^2]} \frac{p^+}{(k^+ - p^+)} \\
& \left[\delta_{ik}\,\delta_{jm} - \frac{k^+}{p^+}\,\delta_{jk}\,\delta_{im} - \frac{k^+}{(k^+ - p^+)}\,\delta_{ij}\,\delta_{km}\right] \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \\
& \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \Bigg\}
\end{aligned}
\tag{1.8}
$$

Integrating wrt p^+ and we write it in three parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& \frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}+\frac{1}{k^+-\vee}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{k^+}-1\right)+\log\left(\frac{k^+}{\Lambda}\right)\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[\vee\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[(\Lambda+k^+)\,(x-z)^2-k^+\,(x-w)^2\right]}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{\Lambda}{k^+-\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[\Lambda\,(x-z)^2-k^+\,(x-w)^2\right]}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{k^+}\right]+\log\left[\frac{k^+}{\Lambda}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\left[k^+\,(x-w)^2-\vee\,(x-z)^2\right]}{\left[k^+\,(x-w)^2-\Lambda\,(x-z)^2\right]}\right] \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}
\end{aligned}
\tag{1.9}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,[(x-w)^2-(x-z)^2]}\left[\frac{1}{(\vee-k^+)}-\frac{1}{\Lambda}\right] \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{\vee}{\Lambda+k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{1.10}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& \frac{ig^4 f^{abd}}{8\pi^5}\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{k^+-\Lambda}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{k^+-\Lambda}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{k^+-\Lambda}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\Bigg\}
\end{aligned}
\tag{1.11}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& +\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(z-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{1.12}
$$

where

$$
\begin{aligned}
\mathcal{A} = \frac{1}{\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\Bigg\{ & \arctan\left[\frac{(k^+-\Lambda)(x-z)^2 - k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right] \\
& -\arctan\left[\frac{\Lambda(x-z)^2 - k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right]\Bigg\}
\end{aligned}
$$

1.4 Row IV

[Done ✓]

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int_{\Lambda}^{\vee-k^+}\frac{dp^+}{2\pi}\int_z & \Big[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\Big]\times \\
& \Big[U^{ac}(w)U^{bd}(z)\bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+,-z)\mathbb{B}_{2ji}^{dc}(k^+,-w;p^+,-z) - \bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+,-z)\bar{\mathbb{B}}_{2ji}^{ba}(k^+,-w;\,p^+,-z)\Big]
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{4\pi^4}\int_{\Lambda}^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,z,w,x',w'} & e^{-ik(w'-w)}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[p^+(x-z)^2 + k^+(x-w)^2]} \\
& \left[\frac{\delta^{ij}}{2\,(z-w)^2\,(p^+ + k^+)}\left[(x-z)^2 - (x-w)^2\right] + \frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2} - \frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]\right. \\
& \left. + \frac{1}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \\
& \Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]
\end{aligned}
\tag{1.13}
$$

Integrating w.r.t. p^+

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{ & \\
& \frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(\vee-k^+)\,(x-z)^2 + k^+(x-w)^2}{\Lambda\,(x-z)^2 + k^+(x-w)^2}\right]\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
& \left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] \\
& -\frac{ig^4}{16\pi^5}\frac{1}{k^+}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times\log\left[\frac{\vee}{\Lambda+k^+}\right] \\
& +\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\left[\log\left(\frac{k^+}{\Lambda}\right) + \log\left(\frac{\vee}{k^+}-1\right)\right]\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]
\end{aligned}
\tag{1.14}
$$

1.5 Row V [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \int \frac{dp^+}{2\pi} \int_{x,z} \Big[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w') U^{b'a}(w')\Big]\times$$
$$\Big[-\bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+,w-z;p^+,z-x)\bar{\mathbb{B}}^{bc}_{2jk}(k^+-p^+,-z;p^+,-x) + \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+,w-z;p^+,z-x)\,U^{db}(x)\,U^{ce}(z)$$
$$\mathbb{B}^{be}_{2jk}(k^+-p^+,-z;p^+,-x)\Big]$$

We write it in two parts:

1.5.1 First part: [Done ✓]

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{$$
$$\frac{ig^4}{16\pi^4}\frac{1}{k^{+\,2}}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',y'} \frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\delta^{(2)}\Big[w-z+\frac{p^+}{k^+}(z-x)\Big]$$
$$\Big[\delta_{im}\,\delta_{jk} - \frac{k^+}{p^+}\,\delta_{jm}\,\delta_{ik} - \frac{k^+}{(k^+-p^+)}\,\delta_{ij}\,\delta_{km}\Big]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times$$
$$\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]\Bigg\} \tag{1.15}$$

Integrating w.r.t. w, we get

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\Bigg\{$$
$$\frac{ig^4}{16\pi^4}\frac{1}{k^{+\,2}}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\Big[\delta_{im}\,\delta_{jk} - \frac{k^+}{p^+}\,\delta_{jm}\,\delta_{ik} - \frac{k^+}{(k^+-p^+)}\,\delta_{ij}\,\delta_{km}\Big]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times$$
$$\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]\Bigg\} \tag{1.16}$$

Using + prescription, we get

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^4}\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\Big[\delta_{im}\,\delta_{jk} - \Big[\frac{k^+}{p^+}\Big]_+\,\delta_{jm}\,\delta_{ik}\Big]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times$$
$$\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^4}\frac{1}{k^+}\int_0^{k^+}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{dq^+}{2\pi}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\Big[\frac{1}{q^+}\Big]_+$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+-\Lambda}{\Lambda}\Big]\,\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+-\Lambda}{\Lambda}\Big]\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big] \tag{1.17}$$

1.5.2 Second part:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik\cdot(w'-w)}\Bigg\{$$
$$\frac{g^4}{8\pi^4}\,\frac{1}{k^{+\,3}}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{p^+\,(k^+-p^+)}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\delta^{(2)}\left[w-z+\frac{p^+}{k^+}\,(z-x)\right]\times$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]-\frac{k^{+\,2}}{(k^+-p^+)^2}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{k^+\,\delta_{im}}{p^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]-\frac{k^{+\,2}}{p^{+\,2}}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{k^+\,\delta_{im}}{(k^+-p^+)}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\frac{k^{+\,2}}{p^+\,(k^+-p^+)}\,\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[f^{adc}\,f^{ebf}\,U^{db}(x)\,U^{ce}(z)-f^{abc}\,f^{cbd}\,U^{df}(y)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \tag{1.18}$$

Integrating w.r.t. w, we get

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\int_{x,y,z,x'}\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[f^{adc}\,f^{ebf}\,U^{db}(x)\,U^{ce}(z)-f^{abc}\,f^{cbd}\,U^{df}(y)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \tag{1.19}$$

This can be written as:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c U^{af}(y)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \tag{1.20}$$

This term contains a UV divergence at $x\to z$. So, we use the following identity to isolate the UV term and the finite term. The identity as follows:

$$f^{abc}\,f^{def}\,U^{be}(z)\,U^{cf}(z)=N_c\,U^{ad}(z)$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(y)+N_c\,U^{af}(x)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.21}$$

This can be written in two parts:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\left\{\right.$$
$$\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\right\}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\Big\} \tag{1.22}$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.23}$$

We will handle the p^+, using + prescription for the first part. We get the following:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\Bigg\}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$+\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.$$
$$\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}$$
$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.24}$$

For the second part, we will first handle the UV divergence and then do plus prescription to handle the p^+.

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} + \frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k}$$

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\left[2\log\left(\frac{\vee}{\Lambda}\right)-2\log\left(\frac{\vee}{k^+}\right)+2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}+\frac{4\Lambda}{k^+}-\frac{\Lambda^2}{k^{+2}}+\frac{2\Lambda^3}{3k^{+3}}\right]$$
$$\times\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{1.25}$$

Setting $\Lambda\to 0$, we get

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\left[\log\left(\frac{\vee}{k^+}\right)+\log\left(\frac{k^+}{\Lambda}\right)-2\log\left(\frac{\vee}{k^+}\right)2\log\left(\frac{k^+}{\Lambda}\right)+2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}\right]\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{1.26}$$

$$\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{$$
$$e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}-\frac{p^+}{k^+-p^+}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]$$
$$-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]$$
$$-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]$$
$$\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}$$
$$\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\}. \tag{1.27}$$

Simplifying, we get

$$\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{$$
$$e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]$$
$$-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]$$
$$-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]$$
$$\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}$$
$$\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\}$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}$$
$$\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\frac{k^+}{q^+}\times$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}$$
$$\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.28}$$

Now, we do plus prescription for $\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k}$ and integrating wrt p^+ and setting Λ to be 0 wherever possible. We get the following

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\;\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^{0}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}+2\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\right]\\
&\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.29}
$$

Now, we write the final answer for Row V:

First part

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\\
&\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+\\
&\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\\
&-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\\
&-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]
\end{aligned}
\tag{1.30}
$$

Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.\\
&\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}\\
&\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\,\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.31}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)-\log\left(\frac{\vee}{k^+}\right)2\log\left(\frac{k^+}{\Lambda}\right)+2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}\right]\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\\
&\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y).
\end{aligned}
\tag{1.32}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^\Lambda\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}+2\log\left[\frac{k^+-\Lambda}{\Lambda}\right]\right]\\
&\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.33}
$$